Pankaj Rana

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Nationality Indian (Student Visa)
D.O.B 14 September, 1996

LinkedIn

LeetCode

Professional Experience

May 2025 – Feb. 2026

Master's Thesis Student at Zentrum für Sonnenenergie- und Wasserstoff-Forschung Baden-Württemberg (ZSW)

Project Real-time Bat detection on wind turbines | 3D Object detection with AI

- Processed large-scale LiDAR recordings to extract temporally consistent 3D object trajectories using Kalman-based multi-object tracking.
- Developed supervised trajectory embedding models (TCN backbone with supervised contrastive learning) on synthetic motion datasets.
- Designed unsupervised autoencoder architectures for centroid-based trajectory representation learning.
- Implemented spatio-temporal BEV encodings and trained 3D U-Net-based models for voxel-wise occupancy prediction under extreme class imbalance.
- Evaluated models using IoU, precision, recall, F1-score, and latent space visualization (t-SNE).

Aug. 2024 – Feb. 2026

Research Assistant at the University of Duisburg-Essen

Computer Science, DSA, and Programming

- Conceptualized and developed programming exercises in Data Structures and Algorithms (DSA).
- Implemented solutions in Python, emphasizing algorithmic efficiency, scalability, and clean coding practices.
- Designed and executed pytest test frameworks to validate algorithmic problem solutions, ensuring correctness across diverse edge cases.

Nov. 2024 – April 2025

Software Development Intern at Porsche Engineering Services

Project ADAS Training Simulator | Software Development and Autonomous Driving

- Developed a Windows application in C++ to control the steering wheel, seat actuator, and other hardware components of the ADAS training simulator.
- Implementation of control loops for precise control of movements based on real-time sensor data in the CARLA simulation environment.
- Focus on control engineering, real-time processing and hardware integration.

June 2024 - Sept. 2024

Research Assistant at DST - Development Center for Ship Technology and Transport Systems e.V.

Project Cobotank | Robotics

- Implemented transformation from AprilTag coordinate system to the robot base frame in ROS 2 with C++ and Python.
- Creation and optimization of the Octomap and combination of the camera model with the Cobotank model.
- Carrying out error diagnoses and fixes as well as managing version control.

Education

April 2022 - Present

Master's Degree (M.Sc.) University of Duisburg-Essen, Germany

Mechanical Engineering (Specialization in Mechatronics)

- Machine Learning & Artificial Intelligence
- Data Analysis
- Robotics
- Control Systems
- Current grade: 2.1

Projects

April 2024 – Oct. 2024

Traffic and Battery status analysis using Machine Learning

University of Duisburg-Essen

- **Traffic data classification:** Implemented and compared machine learning classifiers (e.g., Decision Trees, Random Forests) for speed analysis and flow category classification using Python with scikit-learn and Pandas.
- **Battery health prediction:** Predicted battery State of Health (SoH) and Remaining Useful Life (RUL) using the NASA Li-ion dataset, leveraging Python, NumPy, and scikit-learn for data processing, feature engineering, and model development.
- Gained experience in data preprocessing, supervised learning, and performance benchmarking.

Oct. 2023 - Jan 2024

Robotics Application - Board Plotter

University of Duisburg-Essen

- Programmed the ESP32 microcontroller in C++ to control a board-mounted cable plotter.
- Development and implementation of precise motion logic and algorithms.
- Configuration, calibration and ensuring the accuracy of the motors.
- Integration of safety mechanisms and error handling.
- Carrying out tests and optimizations to improve performance.

Aug. 2016 - July 2020

Bachelor of Engineering (B.E.) Panjab University, India

Mechanical Engineering

- Control Systems
- Programming Fundamentals
- Final grade: 2.25

Skills

Language Skills

German (B2), **English** (Fluent: C2)

Programming Languages

- Python
- C++

Machine Learning & AI Frameworks

- PyTorch, TensorFlow, Keras
- pytest
- scikit-learn, NumPy
- OpenCV, Open3D
- Matplotlib
- Jupyter Notebook
- Slurm

Data Processing & Tools

- Data Preprocessing, Feature Engineering, Model Evaluation
- GitHub/ GitLab
- Docker (Basics)
- Linux

Development Methodologies

- V-Modell
- Agile (Scrum, Kanban)